MA3K7 Assignment 2

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1 Entry Phase

We play on an $n \times n$ grid, which we interpret as an $n \times n$ matrix $A = (a_{ij})$. Player 0 moves first and writes a 0 in an empty cell; Player 1 writes a 1 in an empty cell. They alternate until the matrix is completely filled. Player 0 wins iff $\det(A) = 0$.

Our aim is to investigate the game and, in particular, to find a strategy for Player 0 that forces $\det(A) = 0$ regardless of Player 1's moves.

Investigation plan.

- **I know:** A is an $n \times n$ matrix with entries in $\{0, 1\}$; Player 0 moves first; Player 0 wins iff $\det(A) = 0$.
- **I want:** determine whether Player 0 has a forced win, and (if so) describe an explicit strategy.
- **I introduce:** determinants as a test of linear dependence of rows; it is enough to force a non-trivial linear relation.
- **I try first:** analyse small cases $n = 1, 2, 3$, look for patterns, then generalise to $n \geq 4$.

2 Attack Phase

2.1 $n = 1$

If $n = 1$, then Player 0 writes 0 in the only cell, so $A = [0]$ and $\det(A) = 0$. Hence Player 0 wins.

2.2 $n = 2$

Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{with} \quad a, b, c, d \in \{0, 1\}.$$

Then

$$\det(A) = ad - bc.$$

In a 2×2 matrix with entries in $\{0, 1\}$, we have $\det(A) = ad - bc \in \{-1, 0, 1\}$. Moreover, $\det(A) \neq 0$ if and only if $ad \neq bc$, i.e. one of the diagonal products is 1 and the other is 0. Player 0 can always prevent this in the game.

Strategy for $n = 2$. By symmetry, suppose Player 0 starts by placing 0 in the $(1, 1)$-cell, so $a = 0$. Now consider Player 1's first move:

- If Player 1 plays 1 at $(1, 2)$ (so $b = 1$), then Player 0 plays 0 at $(2, 1)$ (so $c = 0$). The remaining cell $(2, 2)$ is then forced to be 1, giving $A = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$ and $\det(A) = 0 \cdot 1 - 1 \cdot 0 = 0$.
- If Player 1 plays 1 at $(2, 1)$, Player 0 responds symmetrically by playing 0 at $(1, 2)$. Again the final determinant is 0.
- If Player 1 plays 1 at $(2, 2)$, then Player 0 plays 0 in either remaining cell. In both cases one of ad or bc is forced to be 0, hence $\det(A) = 0$.

Therefore Player 0 wins for $n = 2$.

2.3 $n = 3$: turning the determinant into "row products minus column products"

For 3×3, expanding the determinant produces six triple-products. A useful relabelling groups these products as three "row products" minus three "column products".

Lemma 1. *Let*

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Define a new 3×3 array

$$B = \begin{pmatrix} a_{11} & a_{22} & a_{33} \\ a_{23} & a_{31} & a_{12} \\ a_{32} & a_{13} & a_{21} \end{pmatrix}.$$

Then

$$\det(A) = \Big(\sum_{\text{rows } r \text{ of } B} \prod_{x \in r} x \Big) - \Big(\sum_{\text{cols } c \text{ of } B} \prod_{x \in c} x \Big),$$

i.e. $\det(A)$ *equals the sum of the three row-products of B minus the sum of the three column-products of B.*

Proof. Expanding $\det(A)$ by the Leibniz formula yields

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

The three positive terms are exactly the products of the three rows of B, and the three negative terms are exactly the products of the three columns of B. $\square$

Sufficient condition for $\det(A) = 0$ when $n = 3$. Since entries are in $\{0, 1\}$, a triple-product is 1 iff all three entries are 1; otherwise it is 0. Thus, if every row and every column of B contains at least one 0, then all six triple-products vanish and hence $\det(A) = 0$. This gives a concrete sufficient condition for $\det(A) = 0$ in the $n = 3$ case and motivates the general idea of "forcing linear dependence" rather than computing determinants directly.

A key dead end and a re-entry. My first instinct was to try to force two identical rows (which would immediately imply $\det(A) = 0$). However, Player 0 cannot place 1's, so "matching" rows is not robust against an adversary. This led me to re-enter from a different viewpoint: rather than forcing equality, I aimed to force a *linear relation* that can be achieved using only the placement of 0's. The simplest such relation is obtained by forcing two pairs of rows to be complementary (summing to the all-ones vector), which automatically yields linear dependence.

2.4 Conjecture

From $n = 1, 2, 3$ and further experiments, I conjectured that Player 0 has a winning strategy for all n. In the next section, I prove this for all $n \geq 4$ by giving an explicit strategy.

3 Solution Phase (general strategy for $n \geq 4$)

3.1 Key idea (how the strategy was found)

A fundamental linear-algebraic observation is

$$\det(A) = 0 \iff \text{the rows of } A \text{ are linearly dependent.}$$

So we do not need to compute the determinant during the game; it is enough to force a non-trivial linear relation among the rows at the end.

The simplest way to get dependence is to make two rows identical, but Player 0 cannot freely create 1's, so enforcing equality of rows is hard. A more robust target is to enforce *complementarity*. Let $u \in \mathbb{R}^n$ be the all-ones vector $u = (1, 1, \ldots, 1)$, and denote the i-th row by R_i. If we can force

$$R_1 + R_2 = u \quad \text{and} \quad R_3 + R_4 = u,$$

then

$$(R_1 + R_2) - (R_3 + R_4) = 0,$$

which is a non-trivial linear dependence among R_1, R_2, R_3, R_4, hence $\det(A) = 0$. Thus, for $n \geq 4$, we focus only on the first four rows and try to force them into two complementary pairs.

3.2 Definitions

Assume $n \geq 4$. Call rows $(1, 2)$ a *complementary row-pair* and rows $(3, 4)$ another. For each column j, the two cells $(1, j)$ and $(2, j)$ form a *complementary cell-pair*, and similarly $(3, j)$ and $(4, j)$ form a complementary cell-pair.

A complementary cell-pair is

- *blank* if both cells are empty,
- *complete* if both cells are filled and their entries sum to 1 (so the pair is $(1, 0)$ or $(0, 1)$),
- *half-open* if exactly one cell is filled with 0 and the other is blank.

Theorem 1. *For every $n \geq 4$, Player 0 has a winning strategy (i.e. can force $\det(A) = 0$).*

Proof. We present a two-phase strategy. (For $n = 4$, the game starts directly in Phase 2 because there are no cells outside the first four rows; note that initially all complementary cell-pairs are blank, so the first move in Phase 2 is always possible.)

Phase 1 (pairing responses, while possible)

Player 0 follows these rules:

1. If Player 1 plays a 1 in one of the first four rows and the complementary cell in that complementary cell-pair is blank, then Player 0 immediately plays 0 in the complementary cell (thus completing that pair with sum 1).
2. If Player 1 plays outside the first four rows, then Player 0 also plays outside the first four rows (in any blank cell), *unless* there are no blank cells outside the first four rows. In that case, switch to Phase 2.
3. In any other situation in Phase 1, Player 0 plays outside the first four rows if possible; otherwise switch to Phase 2.

Invariant (Phase 1). While Phase 1 is being followed, every complementary cell-pair in the first four rows is either *blank* or *complete*. In particular, Phase 1 never creates a half-open pair.

Phase 2 (creating and maintaining exactly one half-open pair)

Phase 2 begins at a moment when there are no blank cells outside the first four rows. By the Phase 1 invariant, at this moment every complementary cell-pair is either blank or complete.

If there are no blank pairs, the first four rows are already fully complete-paired, so Player 0 wins. Otherwise:

1. Player 0 chooses one *blank* complementary cell-pair and plays 0 in one of its cells, creating exactly one *half-open* pair.

2. From now on, Player 0 enforces the Phase 2 invariant:

   At the start of every Player 1 *turn, there is exactly one half-open complementary cell-pair, and every other complementary cell-pair is either blank or complete.*

   To maintain this invariant, after Player 1 moves, Player 0 responds as follows:

   - (Case A) If Player 1 plays in a cell whose complementary cell is blank (so Player 1 created a $(1, \square)$ situation in some pair), then Player 0 plays 0 in that complementary blank cell, completing that pair.
   - (Case B) If Player 1 plays in the blank cell that complements the 0 in the unique half-open pair, then that pair becomes complete. If the game is not yet finished, choose any blank pair and play 0 in one of its cells to create a new unique half-open pair.

Conclusion. When the game ends, every complementary cell-pair in rows $(1, 2)$ and $(3, 4)$ is complete and sums to 1 column-wise. Equivalently,

$$R_1 + R_2 = u \quad \text{and} \quad R_3 + R_4 = u.$$

Therefore $(R_1+R_2)-(R_3+R_4) = 0$ gives a non-trivial linear dependence among the rows of A, so $\det(A) = 0$. Thus Player 0 wins. $\square$

4 Review Phase (computational verification)

To sanity-check the strategy for $n \geq 4$, I implemented it in Python and simulated many complete games against a random Player 1 (i.e. Player 1 chooses uniformly at random among currently empty cells and writes 1). After each simulated game, I checked the forced invariants $R_1+R_2 = u$ and $R_3+R_4 = u$ (equivalently, $a_{1j} + a_{2j} = 1$ and $a_{3j} + a_{4j} = 1$ for all columns j). By the proof in the Solution Phase, these invariants imply $\det(A) = 0$.

Implementation. The code follows the proof structure: it maintains the focus on the first four rows, applies Phase 1 whenever there exist empty cells outside those rows, and switches to Phase 2 once the outside region is exhausted. In Phase 2, the code enforces the invariant that there is exactly one "half-open" complementary pair at the start of each Player 1 turn, so Player 1 is eventually forced to complete all remaining complementary pairs, yielding $R_1 + R_2 = u$ and $R_3 + R_4 = u$.

Reproducibility. All simulations were run with a fixed random seed to make the random opponent's moves reproducible. The implementation and instructions to reproduce the experiments are available here:

`https://github.com/rOOOOOnn/MA3K7-Problem-Solving-with-Python`

(Assignment 2 code is in `assignment2.py`.)

Results. Using seed 1, I ran 10,000 trials for each of $n \in \{7, 10, 15, 53\}$. In all cases there were 0 invariant violations, hence (by the argument above) $\det(A) = 0$ in every run. These simulations do not constitute a proof against an optimal opponent, but they provide a useful consistency check for the logic of the strategy and the implementation.

Reflection. The simulations are most useful for validating that the Phase 2 "unique half-open pair" mechanism behaves as intended. In particular, the invariants are properties of the *final* board configuration rather than properties that must hold mid-game. Once the implementation checks the invariants only at the end of each game, the experiments consistently match the proof.

Remark. No computational verification is needed for $n = 1$ and $n = 2$, since these cases are proved directly in the Attack Phase. The simulations are used only as a sanity check for the general $n \geq 4$ strategy.

5 Further thoughts

Swapping the digits (Player 0 writes 1 and Player 1 writes 0). The same structural idea can be reused: instead of forcing complementary pairs to sum to 1, the player who writes 0 can attempt to force complementary pairs to sum to 0 (i.e. both entries 0), or more generally to force a fixed linear dependence among rows or columns. A complete analysis of the swapped-digit game is left for future work.

Changing the win condition to $\det(A) \neq 0$. If the winning condition were reversed so that Player 0 wins iff $\det(A) \neq 0$, then the interaction changes fundamentally: a strategy like the one proved above (which forces a row/column dependence) becomes a *blocking* strategy. In particular, if Player 1 adopts a strategy that forces $\det(A) = 0$, then Player 0 cannot win under the $\det(A) \neq 0$ rule, because the final determinant is locked to be zero regardless of Player 0's choices. This highlights that forcing $\det(A) = 0$ is "easier" in the sense that it can be certified by a single explicit linear dependence, whereas forcing $\det(A) \neq 0$ typically requires ensuring full rank, which is a global condition and is harder to maintain under adversarial moves.